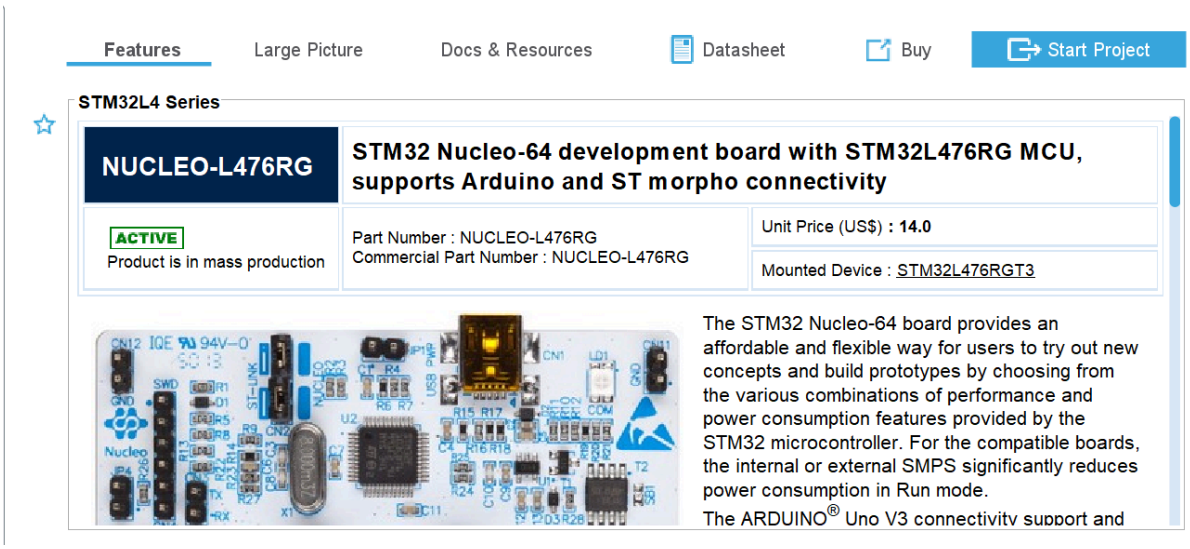
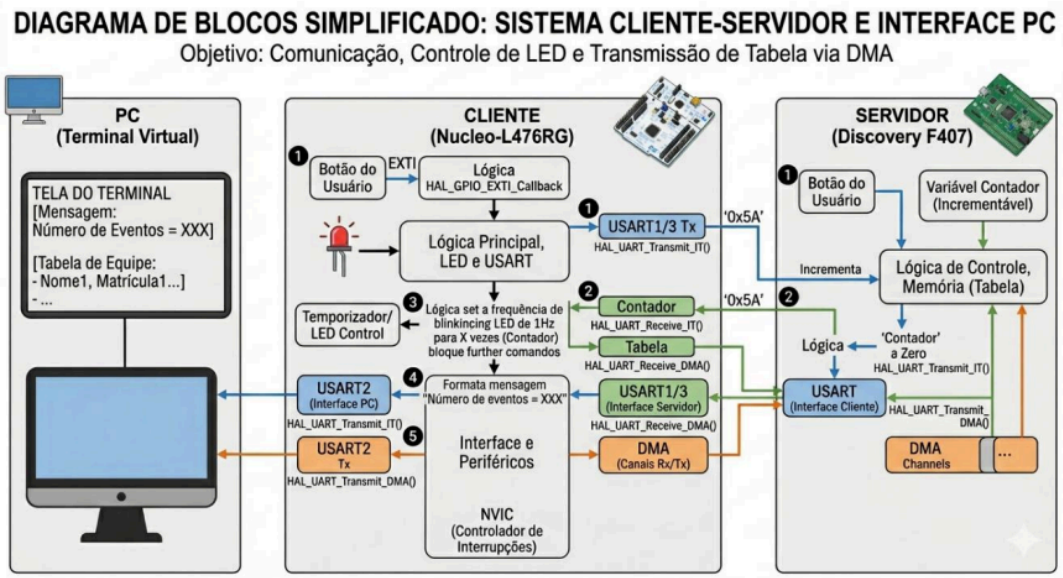
Configuração dos Projetos

Projetos do STM32CUBEmx



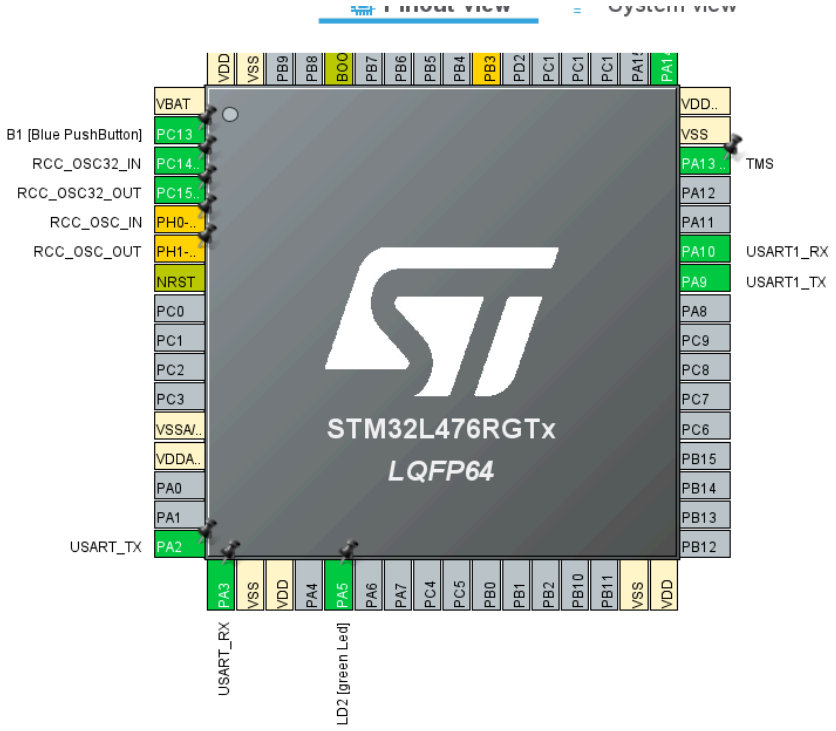
Configuração NUCLEO (L476RG)

O Projeto do professor solicita:

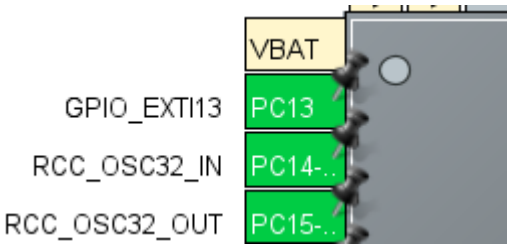


(Imagem gerada por IA)

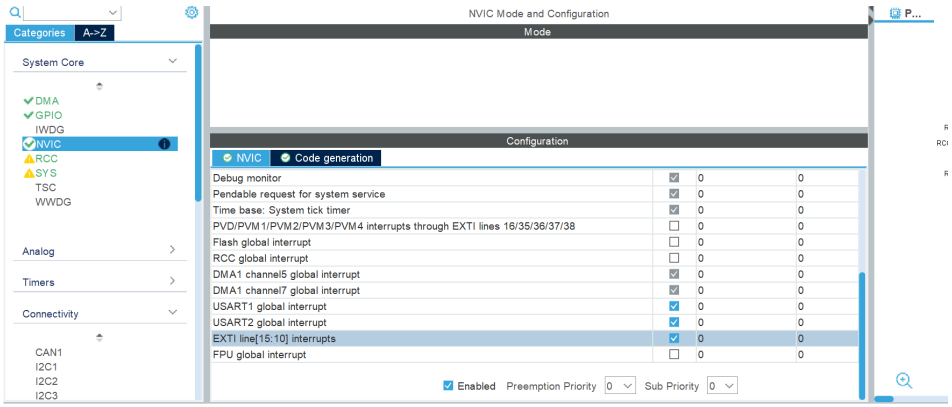
por isso, adotamos as configurações:



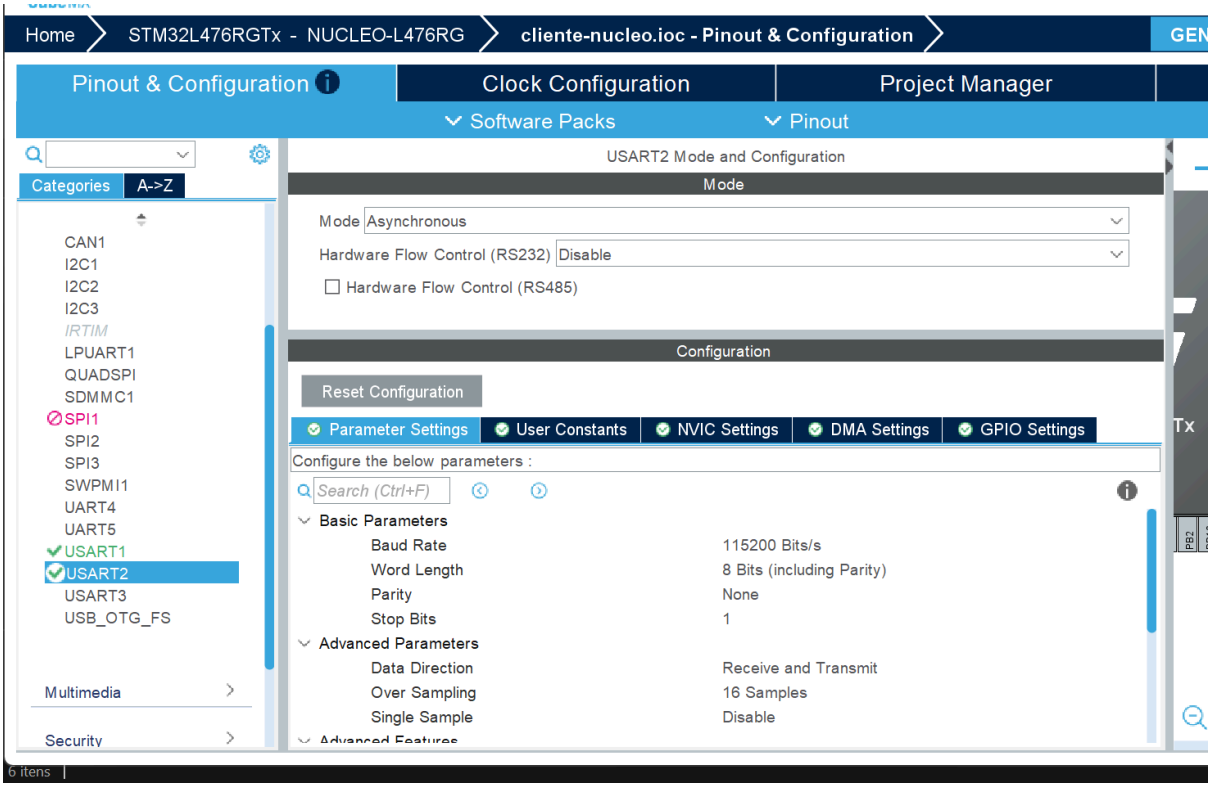
realizamos a substituição do pin PC13

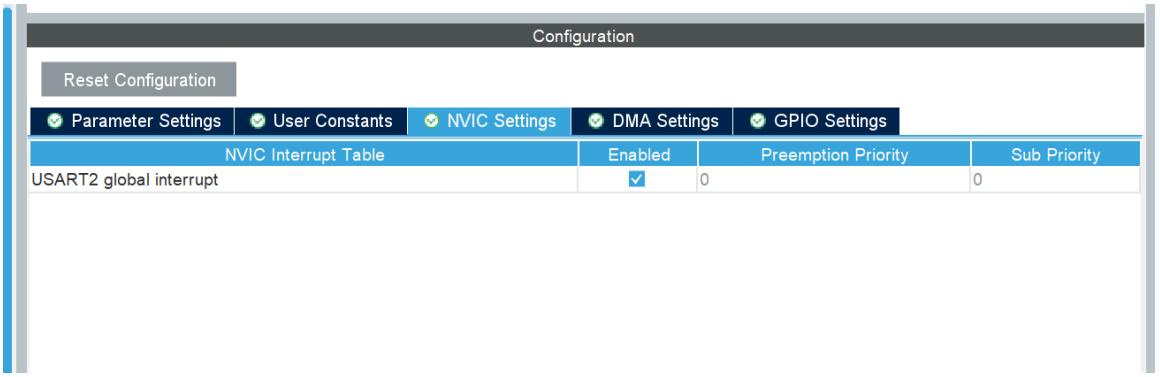


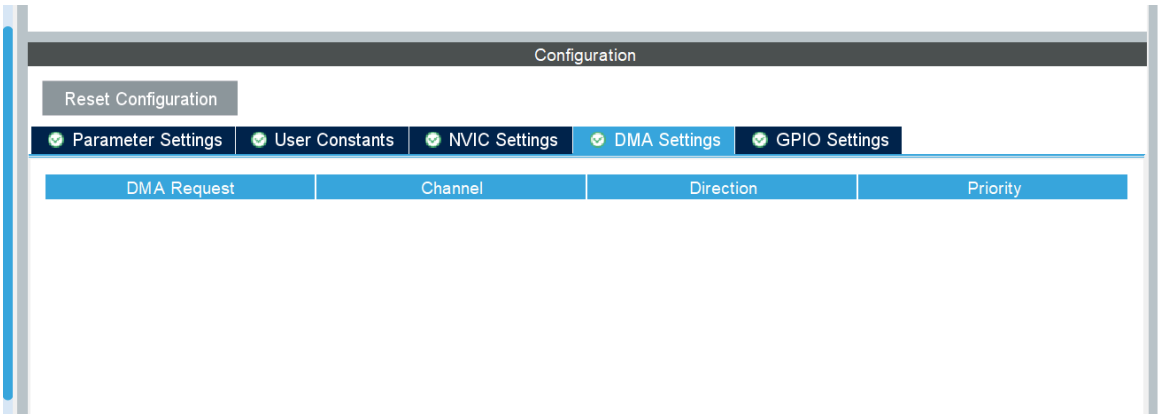
por fim, marquei o NVIC

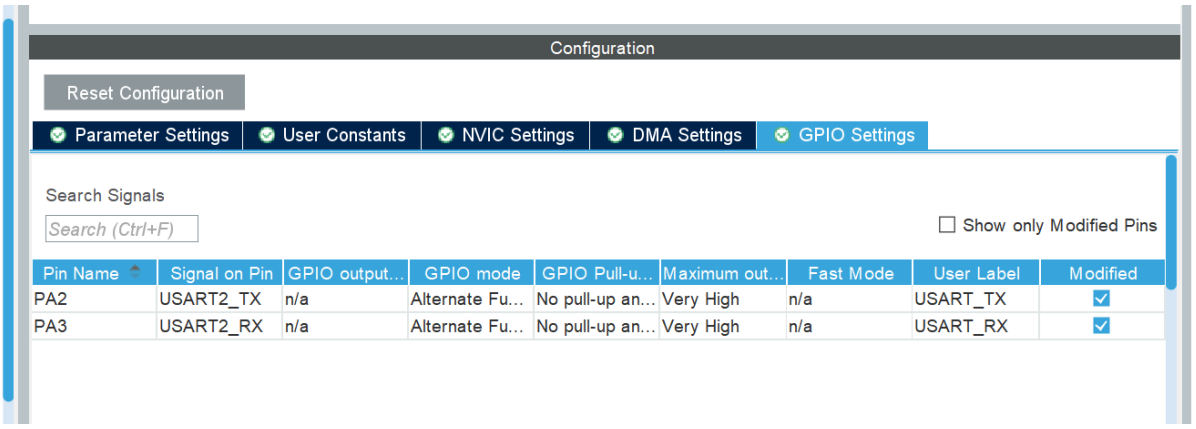


Para USART2, que será ligada com a maquina:

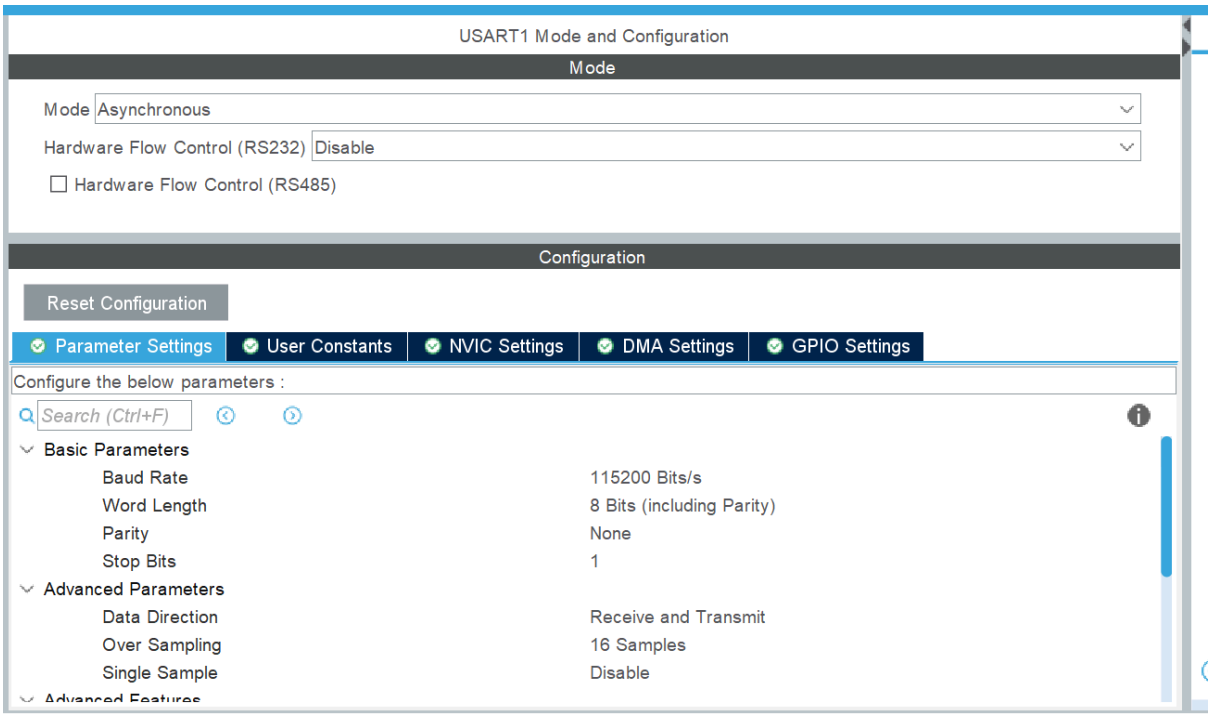


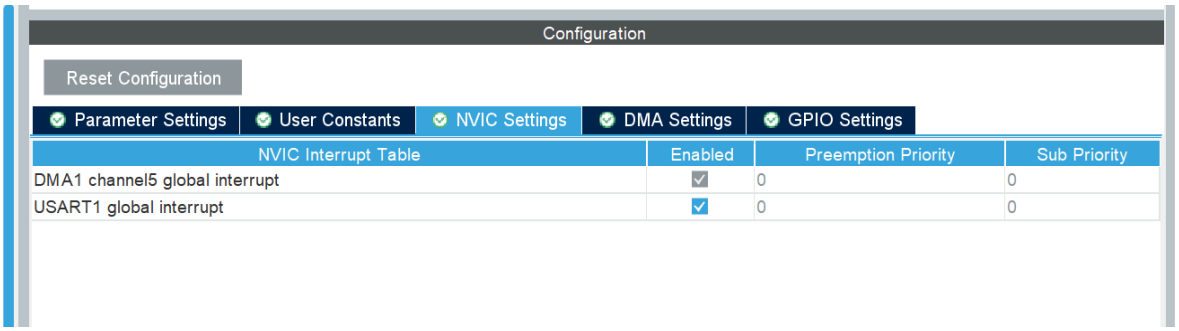






Para USART1, que será ligado na Discovery:





Configuration

Reset Configuration

Parameter Settings | User Constants | NVIC Settings | DMA Settings | GPIO Settings

DMA Request	Channel	Direction	Priority
USART1_RX	DMA1 Channel 5	Peripheral To Memory	Low

Add | Delete

DMA Request Settings

		Peripheral	Memory
Mode	Normal	Increment Address	☐	☑
		Data Width	Byte	Byte

Configuration

Reset Configuration

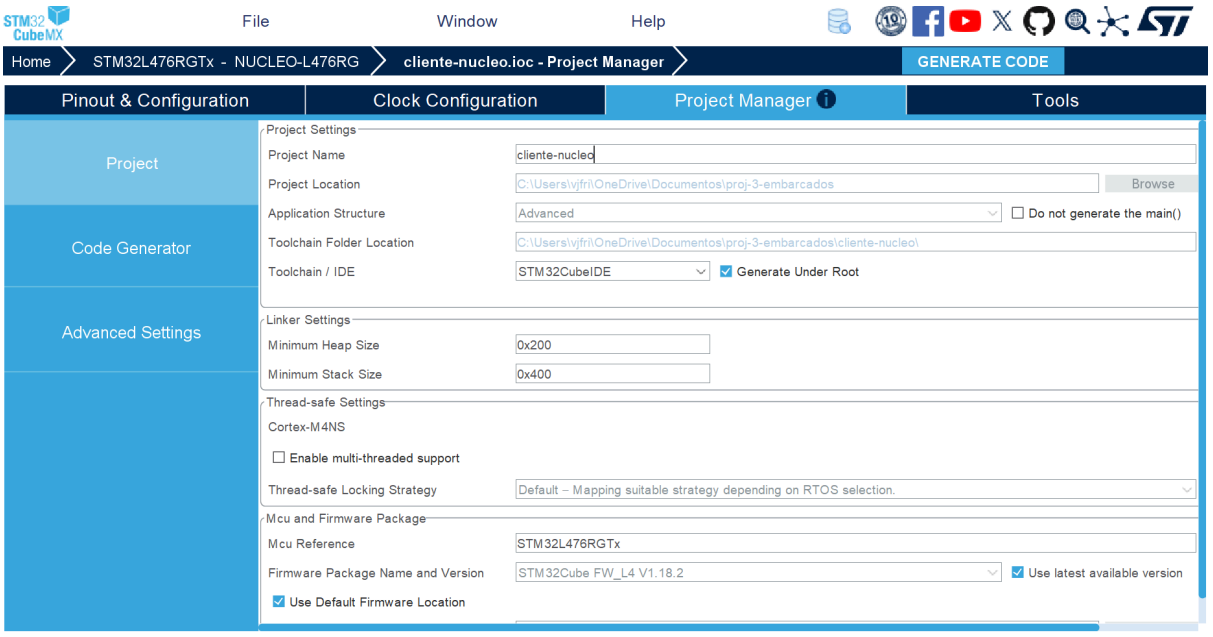
Parameter Settings | User Constants | NVIC Settings | DMA Settings | GPIO Settings

Search Signals

Search (Ctrl+F)

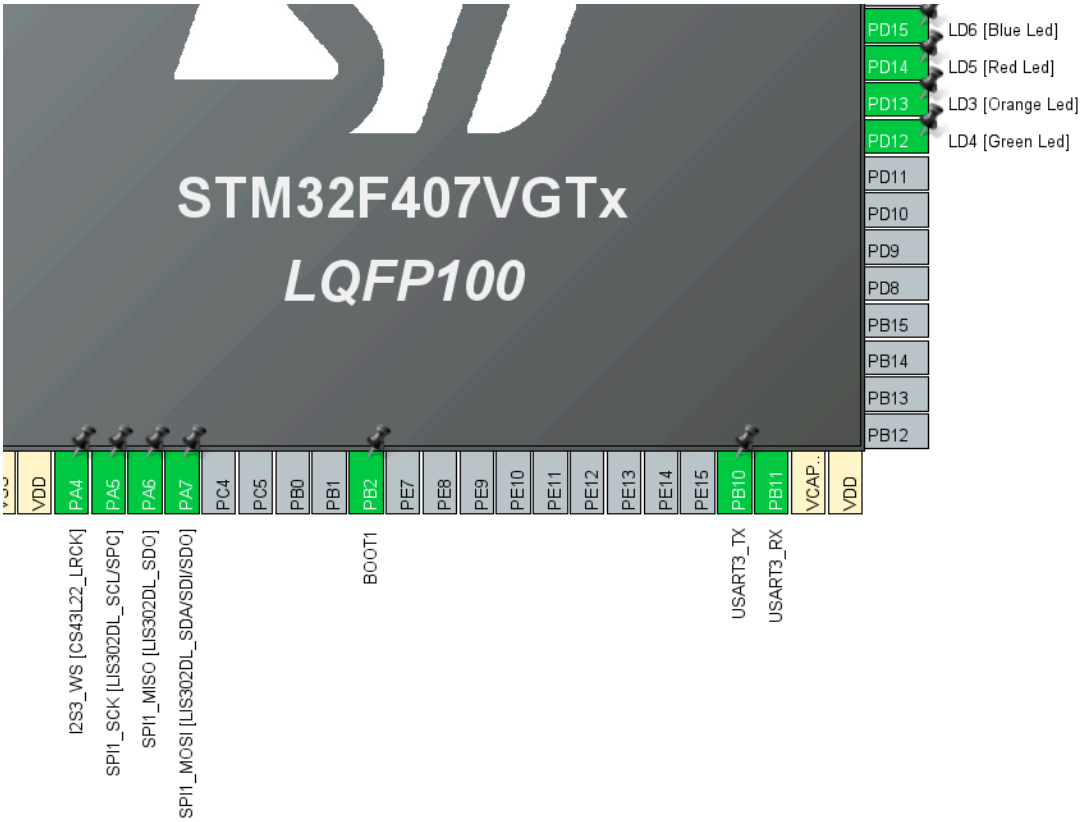
☐ Show only Modified Pins

Pin Name	Signal on Pin	GPIO output...	GPIO mode	GPIO Pull-u...	Maximum out...	Fast Mode	User Label	Modified
PA9	USART1_TX	n/a	Alternate Fu...	No pull-up an...	Very High	n/a		☐
PA10	USART1_RX	n/a	Alternate Fu...	No pull-up an...	Very High	n/a		☐

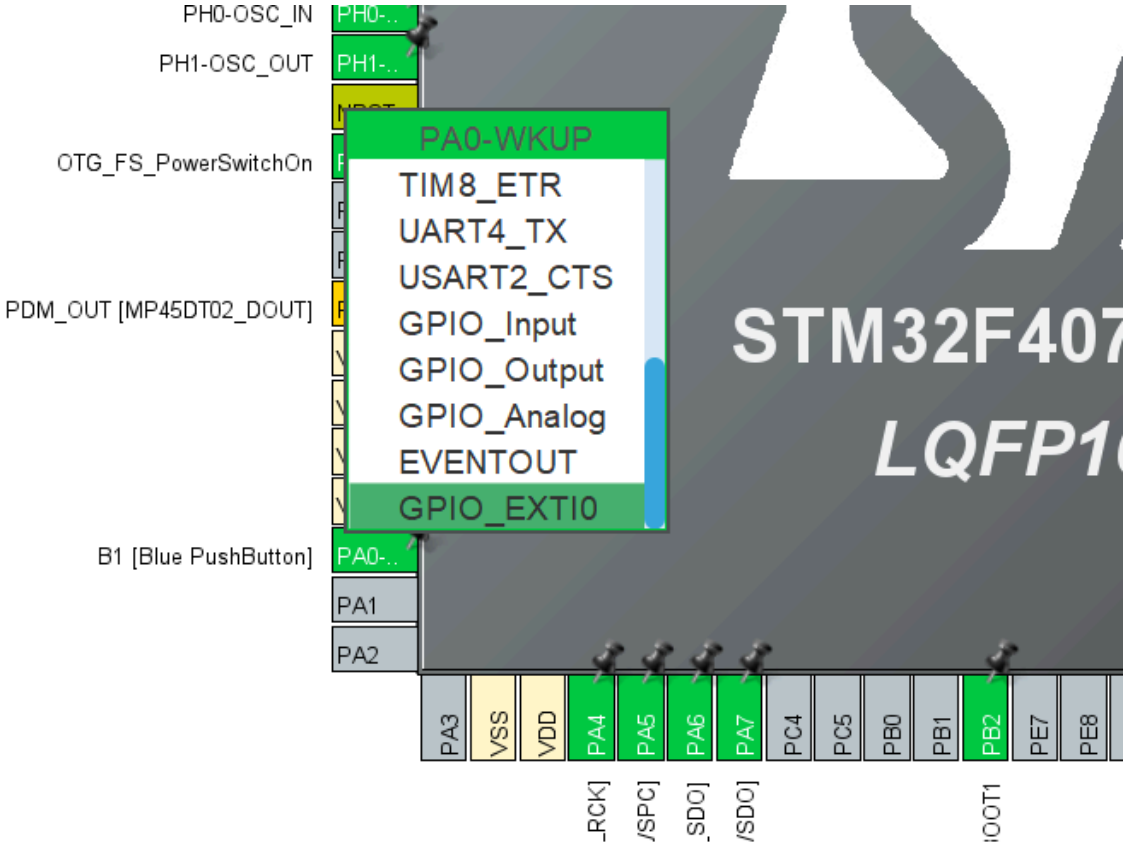


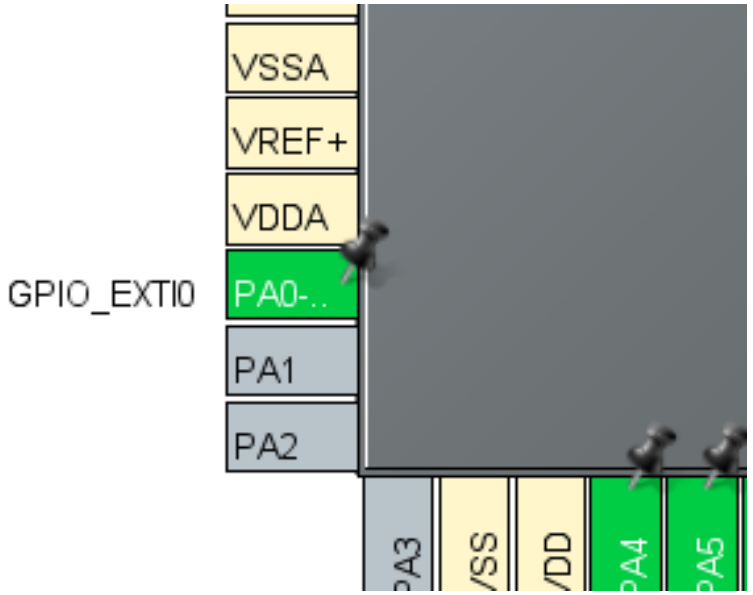
(OBS: sempre gerar código e colocar a IDE correta!)

Configuração DISCOVERY (F407)

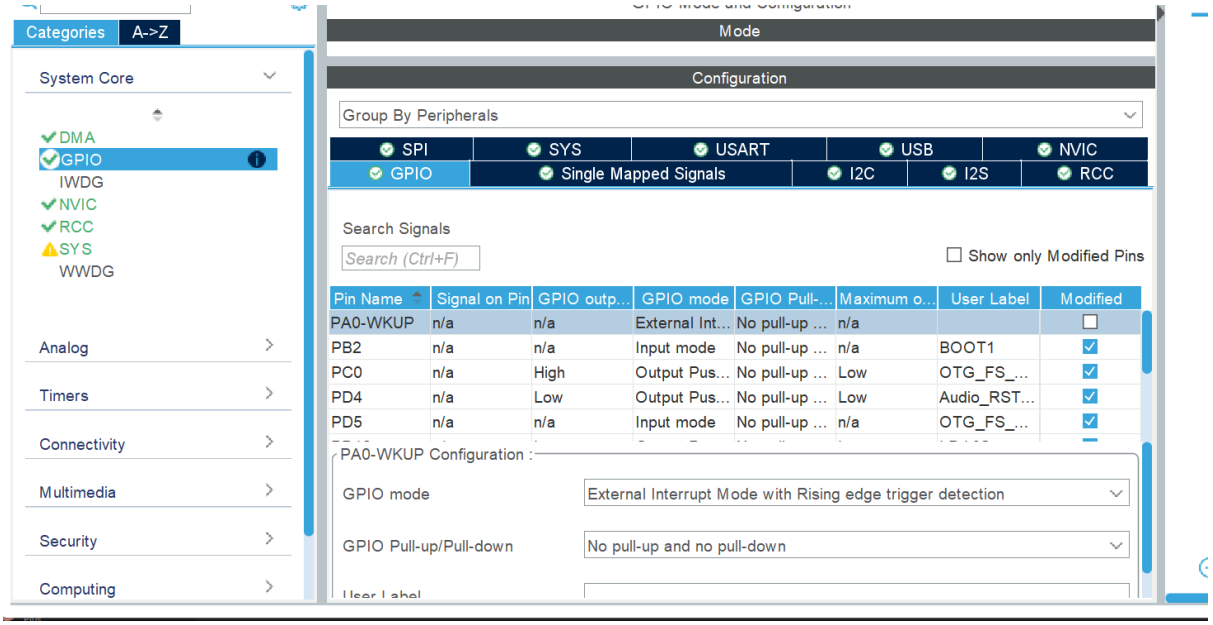


Coloquei o botão para ser o EXTi

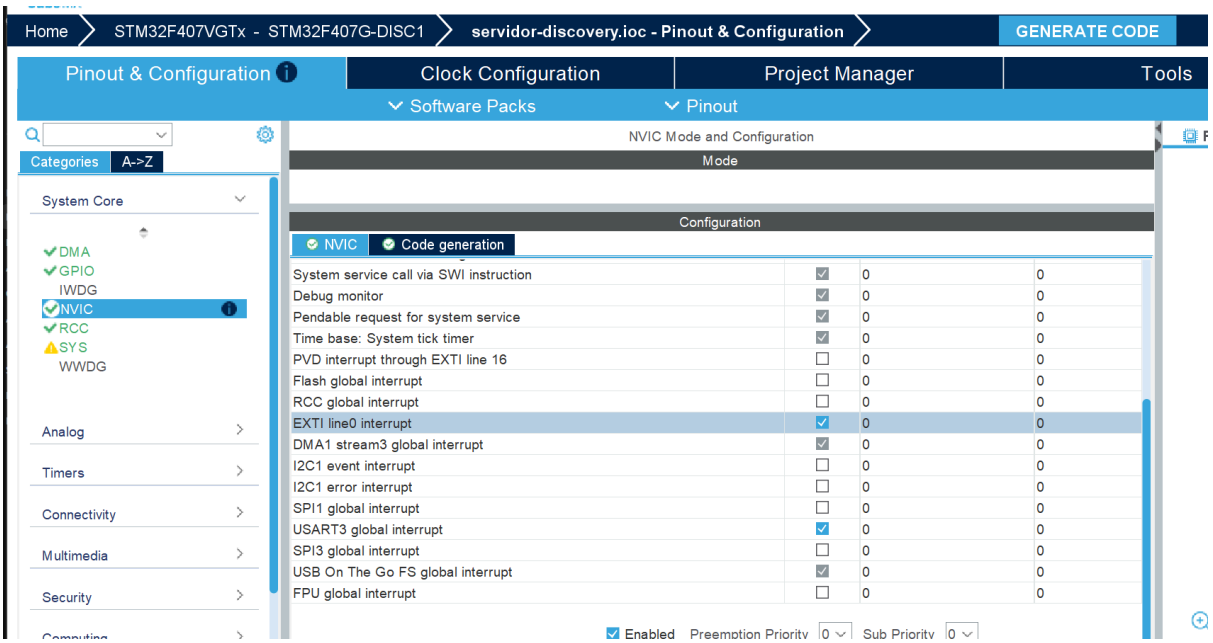




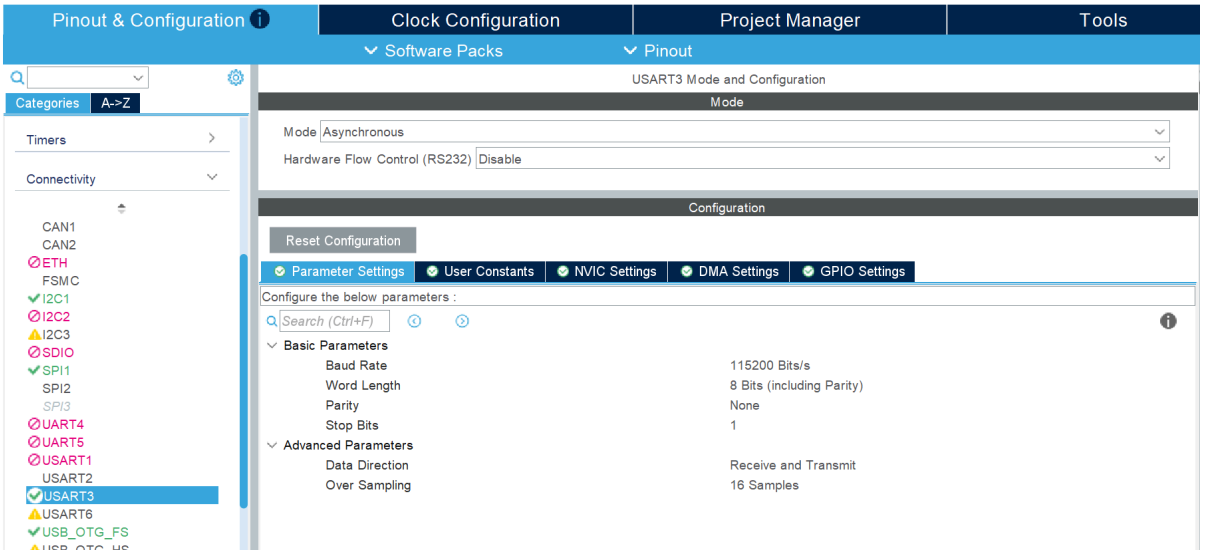
conferimos como o GPIO está:

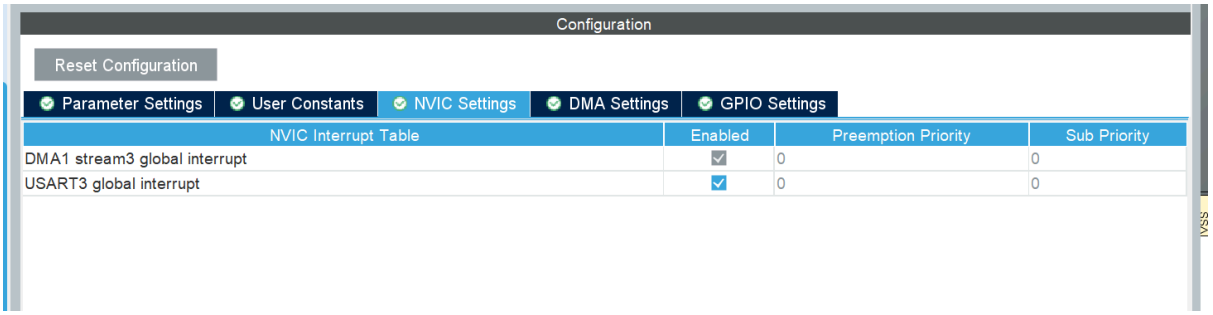


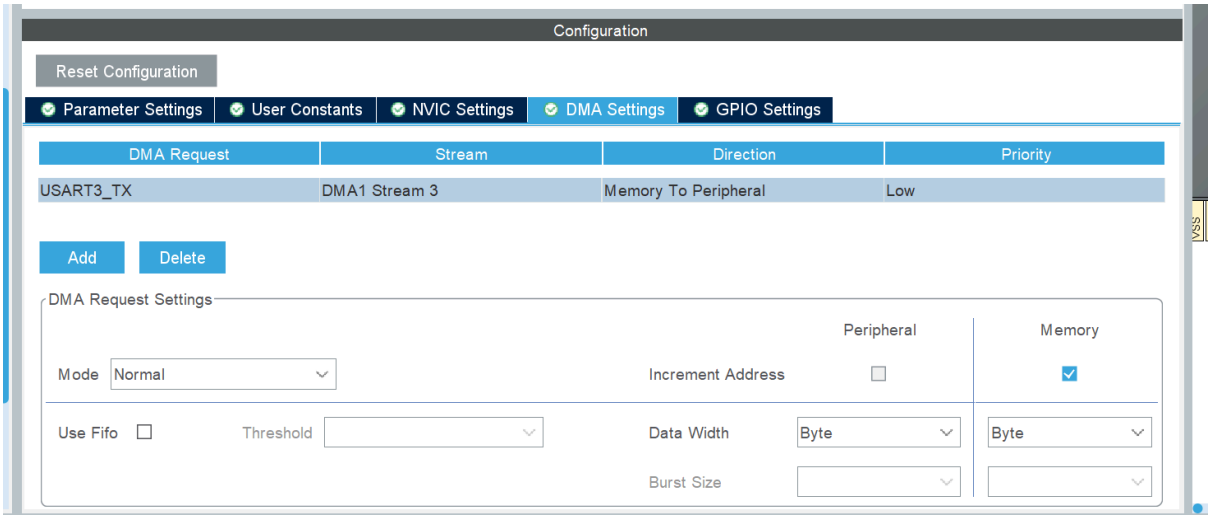
e marcamos a opção do EXTi

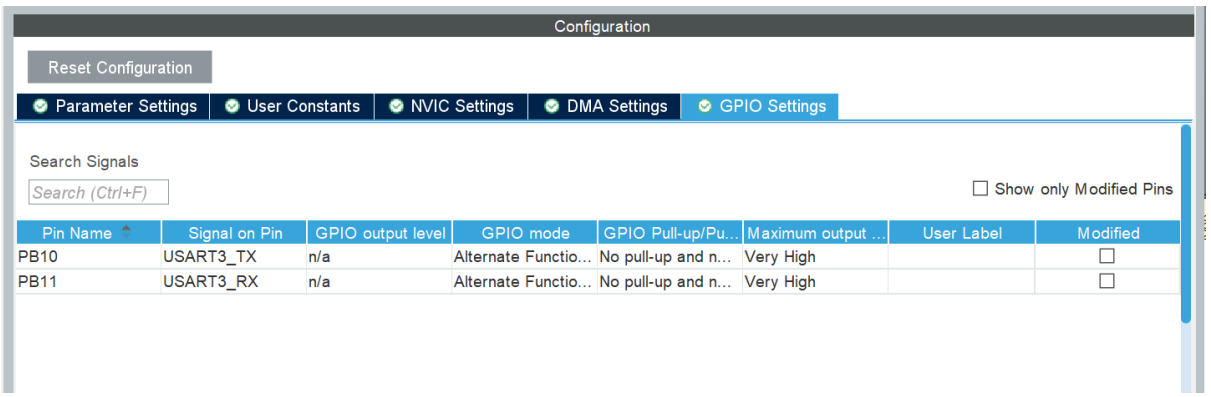


COMUNICAÇÃO:









Ligações físicas

Dados:

Núcleo GPIO D2(PA10) [USART1_RX] — jumper roxo----> [USART3_TX] PB10 Discovery.
Núcleo GPIO D8(PA9) [USART1_TX] — jumper cinza---> [USART3_RX] PB11 Discovery.

Alimentação:

Para gravar: ambas ligadas no USB

Para rodar o Projeto:

- 3v3 da nucleo —jumper marrom—> 3v da Discovery
- GND da Nucleo —jumper preto—> GND da discovery

(seguimos nesse modelo como solicitação o prof. em sala.).

Codigo

Codigo Nucleo

```c
/* USER CODE BEGIN 4 */
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if(GPIO_Pin == GPIO_PIN_13)
    {
        sprintf(log_buffer, "[BTN] Botao pressionado - enviando 0x5A...\r\n");
        log_msg(log_buffer);

        if(estado == IDLE)
        {
            HAL_UART_Transmit_IT(&huart1, &cmd, 1);

            sprintf(log_buffer, "[UART1 TX] Comando enviado.\r\n");
            log_msg(log_buffer);

            estado = WAITING_SERVER;
        }
        else
```

EXTi do botão

```c
        log_msg(log_buffer);
    }
    */

    // reativa recepção do contador
    HAL_UART_Receive_IT(&huart1, &contador_rx, 1);
}

void HAL_UART_RxCpltCallback_DMA(UART_HandleTypeDef *huart)
{
    if(huart->Instance == USART1)
    {
        sprintf(log_buffer, "[DMA RX] Tabela recebida!\r\n");
        log_msg(log_buffer);

        sprintf(log_buffer, "[TABELA RECEBIDA]:\r\n%s\r\n", tabela_rx);
```

reciver do contador na nucleo e Callback do DMA

```c
/* USER CODE BEGIN 3 */
  if(estado == PROCESSING)
  {
      sprintf(log_buffer, "[STATE] -> PROCESSING\r\n");
      log_msg(log_buffer);

      for(int i = 0; i < contador_rx; i++)
      {
          HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
          HAL_Delay(500);
      }

      // garante que termina apagado
      HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);

      sprintf(log_buffer, "[LED] Piscar concluido.\r\n");
      log_msg(log_buffer);
```

blink do led, feito por toggle (daí as piscadas pela metade. é um acendendo e outro apagando)

Codigo Discovery

```c
/* USER CODE BEGIN 4 */
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if(GPIO_Pin == GPIO_PIN_0)
    {
        contador++;

        // só toggle, sem delay
        HAL_GPIO_TogglePin(GPIOD, GPIO_PIN_12);
    }
}

void HAL_UART_RxCpltCallback(UART_HandleTypeDef *huart)
{
    if(huart->Instance == USART3)
    {
        if(rx == 0x5A)
        {
```

codigo e toggle do led.

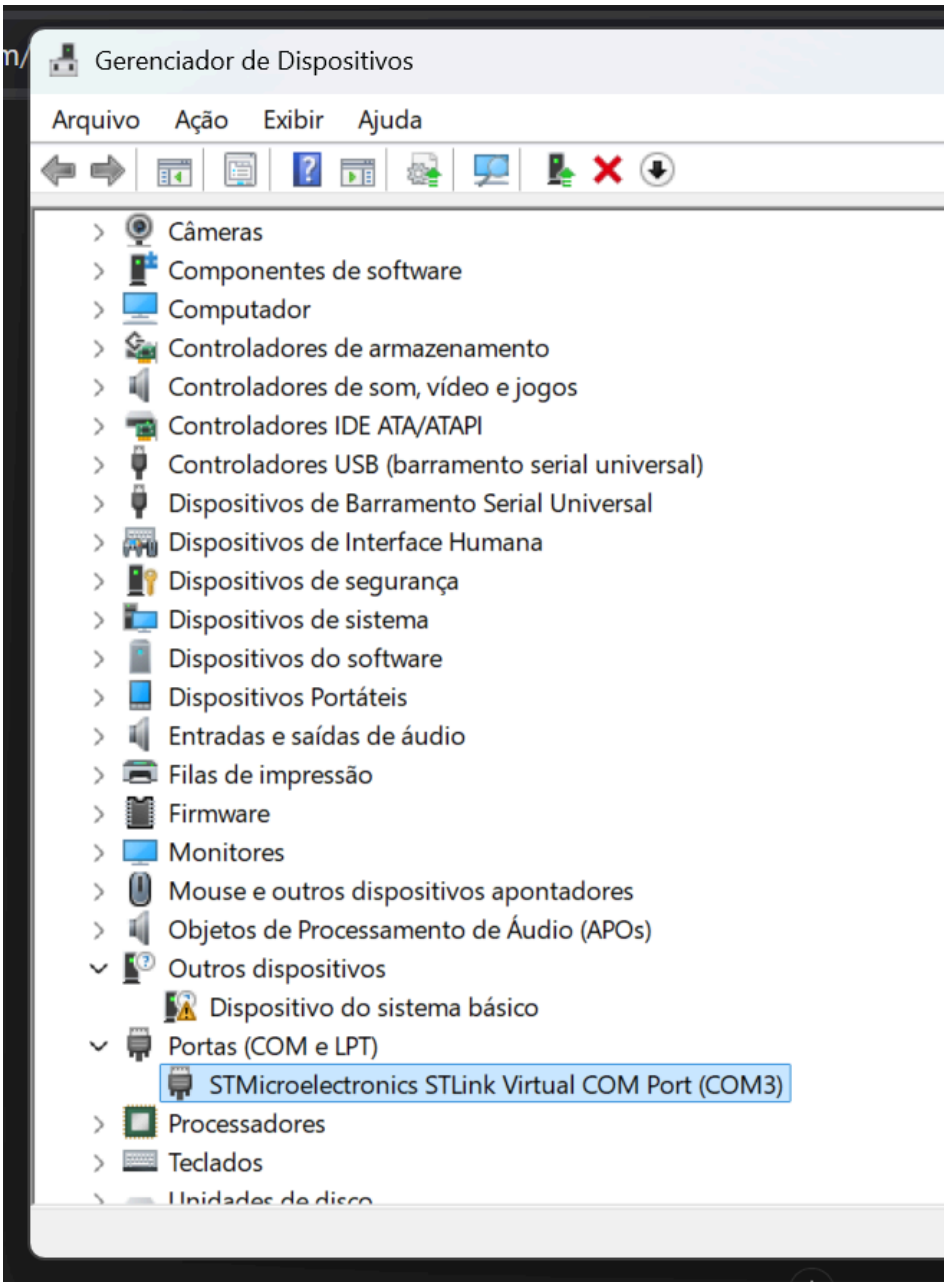
```c
    }
}

void HAL_UART_RxCpltCallback(UART_HandleTypeDef *huart)
{
    if(huart->Instance == USART3)
    {
        if(rx == 0x5A)
        {
            //HAL_UART_Transmit_IT(&huart3, &contador, 1);
            //HAL_UART_Transmit_DMA(&huart3, (uint8_t*)tabela, sizeof(tabela));
            HAL_UART_Transmit(&huart3, &contador, 1, 100);
            HAL_Delay(10); // pequeno delay

            //HAL_UART_Transmit(&huart3, (uint8_t*)tabela, sizeof(tabela), 100);
            uint16_t tamanho = strlen(tabela);
            HAL_UART_Transmit(&huart3, (uint8_t*)&tamanho, 2, 100);
            HAL_UART_Transmit(&huart3, (uint8_t*)tabela, tamanho, 100);
```

recebe a senha e faz o envio do contador e tabela.

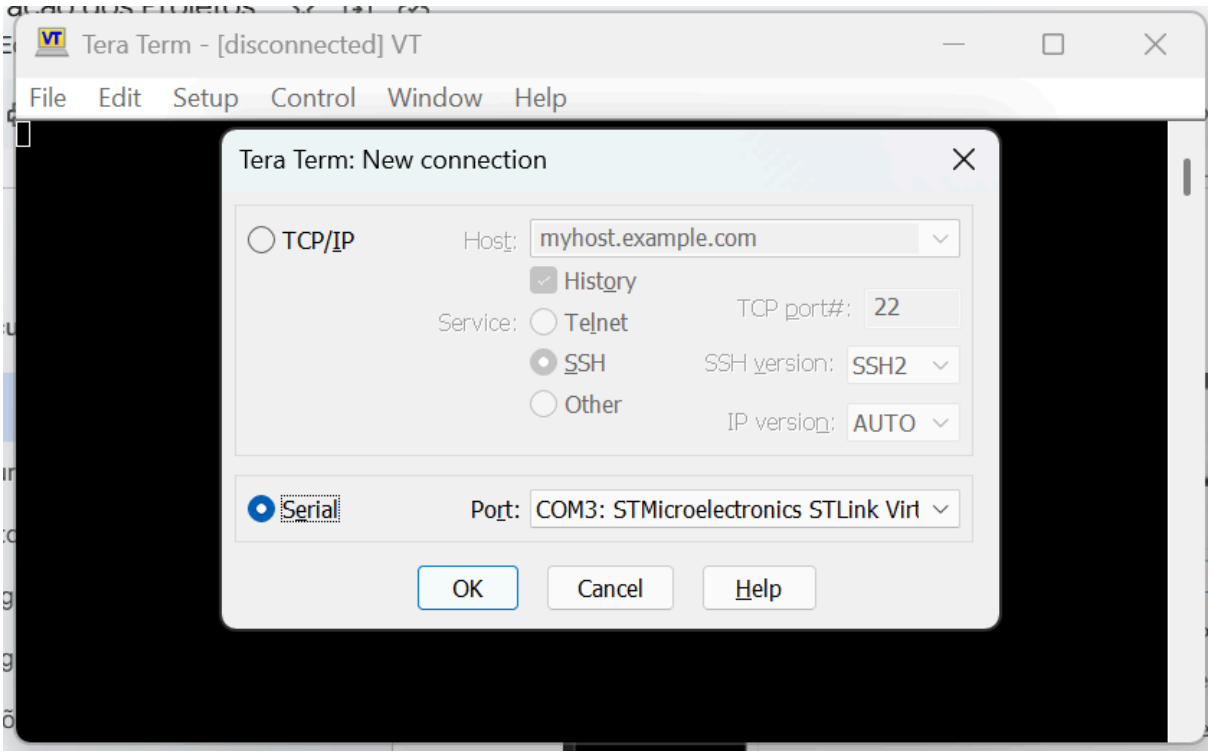
Teste

drivers: https://www.st.com/en/development-tools/stsw-link009.html
Advanced serial port terminal: https://www.com-port-monitoring.com/serial-port-terminal/

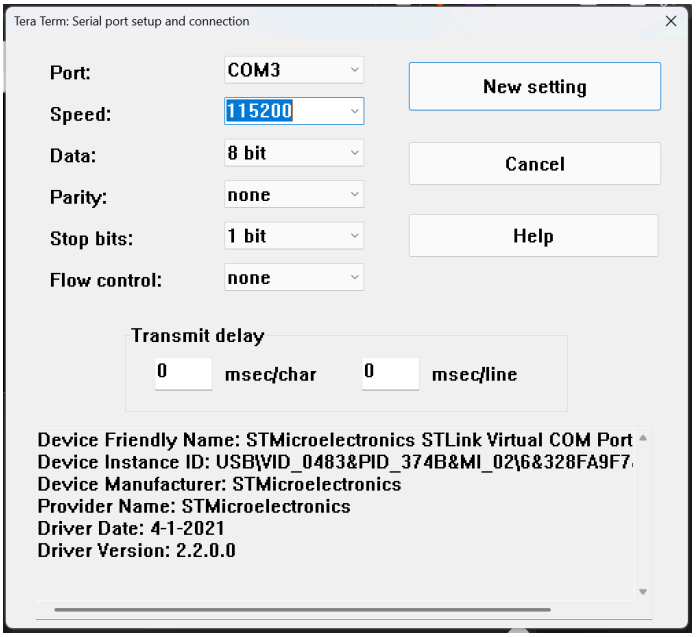
1. Localize a porta COM



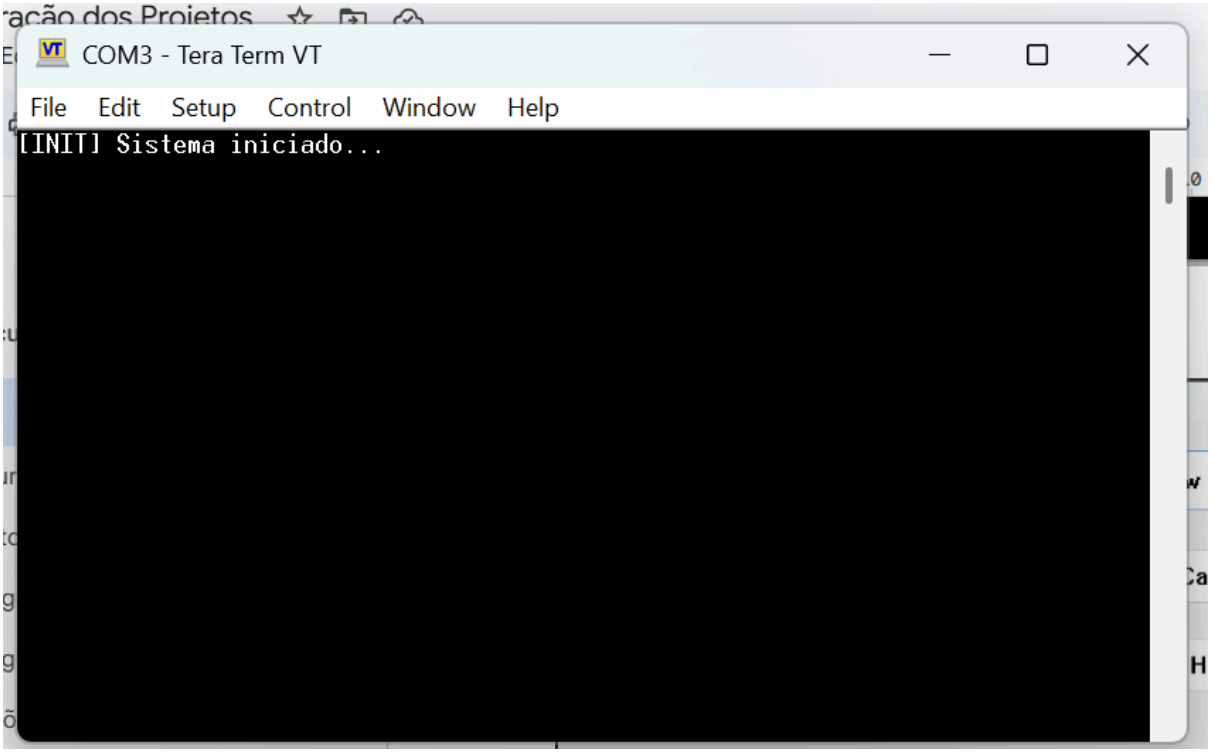
2. Preencha a new session



3. bauld rate // speed



4. codigo de inicialização



5. qnt de botões apertadas recebido

```
[INIT] Sistema iniciado...
[BTN] Botao pressionado - enviando 0x5A...
[UART1 TX] Comando enviado.
[UART1 RX] Contador recebido: 6
[DMA] Recebendo tabela...
[STATE] -> PROCESSING
[LED] Piscar concluido.
Numero de eventos = 6
```